

6 EE PROJECTS STRAIGHT OUT OF A CLASSIFIED LAB

Laser intelligence. Covert communications. Stealth measurement.

Once classified. Now buildable. Still impressive.

1. Laser Surveillance Microphone — hear through a window from 100m, \$15
2. Tap-Proof Laser Communication — encrypted IR link, no RF emission, \$12
3. Laser Rangefinder — 5ns pulse, 100ps TDC, centimeter accuracy, \$25
4. Encrypted Burst Transmitter — 50ms FHSS burst, can't be jammed or DF'd, \$10
5. Passive Night Vision Scope — image amplification with zero emission, \$30
6. Radar Cross-Section Measurement — test stealth materials at your desk, \$12

By @1nska • Electrical Engineer

PROJECT 1

Laser Surveillance Microphone

Hear through a window from 100m — no mic in the room, no RF, no trace

Cost: ~\$15

Difficulty: Advanced

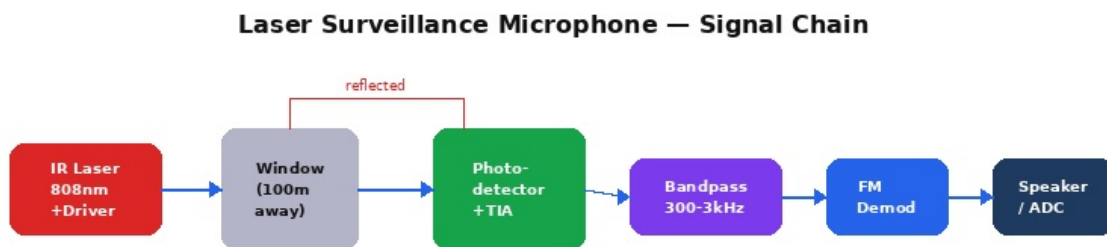
Build Time: 1-2 weekends

Why This Matters

During the Cold War, intelligence agencies developed laser listening devices that could extract audio from vibrating surfaces at distance. The principle is simple: sound waves inside a room cause the window glass to vibrate by nanometers. An IR laser bouncing off that window is phase-modulated by the vibration. A sensitive photodetector reads that modulation, and an FM demodulator extracts the audio. The technique was declassified decades ago, but the analog design challenges remain: sub-millivolt signal detection, laser intensity stabilization, and FM demodulation of an extremely weak carrier.

What this proves to a recruiter: You understand optical signal chains, laser driver design, transimpedance amplification at the noise floor, and analog FM demodulation. These skills directly apply to LIDAR, fiber optics, optical sensing, and defense.

System Architecture



Sound vibrates window glass (nm displacement) → modulates reflected beam
Photodetector reads modulation → FM demod extracts audio → speaker output

How It Works

- **IR laser diode (808nm):** A collimated infrared laser aimed at a window pane. The laser must be intensity-stabilized using an APC (automatic power control) circuit — a monitor photodiode inside the laser package feeds back to the driver, keeping output power constant. Any intensity drift appears as noise in the received signal.
- **Window vibration:** Sound waves inside the room cause the glass to vibrate. A normal conversation at 60dB SPL causes ~1-10nm of displacement on a window pane. This displacement phase-modulates the reflected laser beam — the path length changes by twice the displacement (there and back), creating a Doppler shift proportional to the velocity of the glass surface.
- **Photodetector + TIA:** A large-area silicon photodiode (BPW34 or similar) with a transimpedance amplifier (1M Ω feedback) captures the reflected beam. The DC component represents the average reflected power. The AC component (microvolts) carries the audio information. A coupling capacitor separates them.
- **Bandpass filter (300Hz-3kHz):** A 4th-order Sallen-Key bandpass isolates the speech band. This removes laser noise below 300Hz and high-frequency detector noise above 3kHz.
- **FM demodulator:** The phase modulation from the window vibration is equivalent to FM. A simple PLL-based FM demodulator (LM567 or NE567) or a slope detector (tuned LC circuit on the slope of the resonance curve) extracts the audio. Amplify and output to speaker or ADC.

Key Design Decisions

- **Why 808nm instead of visible laser?** 808nm is invisible to the human eye — the target doesn't know they're being illuminated. It also avoids eye safety concerns at low power levels and matches common, cheap laser diodes.
- **Why APC on the laser driver?** Without intensity stabilization, laser power fluctuations appear as amplitude noise that masks the tiny phase modulation signal. The APC loop must have bandwidth >10kHz to track fast intensity variations.
- **Why FM demodulation, not AM?** The window vibration modulates the phase/frequency of the reflected beam, not the amplitude. Amplitude changes come from beam wander and atmospheric scintillation — these are noise sources, not signal. FM demodulation rejects AM noise inherently.

Bill of Materials

Component	Part / Spec	Qty	~Cost
808nm laser diode	200mW, TO-18 with monitor PD	1	\$2.50
Collimating lens	Glass, 6mm, AR-coated	1	\$1.00
Laser driver	LM317 + APC feedback circuit	1	\$1.00
BPW34 photodiode	Large area, 900nm peak	1	\$1.00
TL074 quad op-amp	TIA + filters + demod	2	\$1.60
LM567 tone decoder	PLL-based FM demod	1	\$0.80
Precision resistors	1% for TIA and filters	~20	\$1.50
Film capacitors	For bandpass filters	~10	\$1.00
Tripod mount	For stable laser aiming	1	\$3.00
Speaker or 3.5mm jack	Audio output	1	\$0.50

Build Steps

Laser transmitter: Mount 808nm laser diode in a collimating lens assembly. Build an APC driver: LM317 as current source, monitor photodiode feedback through an op-amp comparator that adjusts drive current to maintain constant output. Set power to 50-100mW — enough for reflection from 100m but below eye-safety limits for diffuse reflection. Mount on a tripod with fine adjustment screws.

Optical receiver: Large-area BPW34 photodiode with a 10mm focusing lens to collect the reflected beam (which will be spread over a large area at 100m). TIA with $1\text{M}\Omega$ feedback + 1pF compensation cap. Add a narrowband optical filter (800-820nm) in front of the lens to reject ambient light. Shield the entire receiver in a tube to reject stray light from the sides.

Signal conditioning: AC-couple the TIA output through 100nF. First stage: x100 amplification. Second stage: Sallen-Key bandpass 300Hz-3kHz (4th order for sharp rolloff). Verify on oscilloscope: with the laser on a nearby window and someone talking, you should see an AC waveform correlated with the speech.

FM demodulation: Feed the filtered signal into an LM567 PLL configured as an FM demodulator. Set the free-running frequency to match the carrier (center of your bandpass). The demodulated output is a voltage proportional to the instantaneous frequency deviation — this IS the audio signal. Low-pass at 3kHz and amplify for speaker output.

Test and calibrate: Start with a short range test (5m, indoors). Aim laser at a window. Play music inside the room near the window. You should hear the music through your receiver. Increase range gradually — 10m, 20m, 50m, 100m. At each range, adjust the receiver gain and verify SNR. Document the maximum intelligible range and the SNR vs distance curve.

Interview Questions:

- How does sound inside a room modulate a laser beam reflected off the window?
- Why is automatic power control critical for the laser driver?
- What limits the maximum range of this system?
- How does FM demodulation reject amplitude noise from atmospheric effects?
- What countermeasures exist against laser microphone surveillance?
- How would you improve sensitivity — what's the theoretical noise floor?

PROJECT 2

Tap-Proof Laser Communication Link

Encrypted data over invisible IR beam — no RF emission, cannot be intercepted

Cost: ~\$12

Difficulty:
Intermediate-Advanced

Build Time: 1-2 weekends

Why This Matters

Every radio transmission can be intercepted, direction-found, and jammed. Laser communication eliminates all three problems: the beam is directional (impossible to intercept from the side), invisible (808nm IR), and carries no RF signature. Military Special Operations, submarine-to-satellite links, and inter-spacecraft communication all use optical links for exactly these reasons. This project builds a complete encrypted optical link: modulated laser transmitter, avalanche photodiode receiver, clock recovery, and AES encryption.

System Architecture



No RF emission | No spectrum signature | Cannot be intercepted from the side
AES-128 encrypted payload | OOK modulation at 1 Mbps | 200m range

How It Works

- **Transmitter — laser driver with OOK modulation:** An 808nm laser diode driven by a MOSFET switching circuit. On-Off Keying (OOK): laser on = 1, laser off = 0. The MCU (ESP32) encrypts the data with AES-128, adds start/stop framing and Manchester encoding (for clock recovery), then drives the MOSFET at up to 1 Mbps.
- **Receiver — avalanche photodiode (APD):** An APD (or high-speed PIN diode like SFH213) provides internal gain through avalanche multiplication. This gives 10-50x better sensitivity than a standard photodiode. Biased at just below breakdown voltage (typically 100-150V from a boost converter). The received photocurrent feeds a transimpedance amplifier.
- **Clock recovery:** Manchester encoding ensures a transition in every bit period, allowing the receiver's PLL to synchronize its clock to the transmitter's bit rate. Without clock recovery, timing drift over long messages causes bit errors. A simple DPLL (digital PLL) implemented on the ESP32 handles this.
- **AES decryption:** The receiver ESP32 strips framing, decodes Manchester, and decrypts AES-128 using a pre-shared key stored in flash. Output: original plaintext on serial terminal or OLED display.

Bill of Materials

Component	Part / Spec	Qty	~Cost
808nm laser diode	200mW, collimated module	1	\$2.50
SFH213 PIN diode	High-speed, 900nm peak	1	\$1.50
MOSFET	2N7000 (laser switching)	1	\$0.30
ESP32	TX and RX controllers	2	\$6.00
TL072 op-amp	TIA + comparator	1	\$0.80
Boost converter	For APD bias (if using APD)	1	\$2.00
Collimating optics	TX lens + RX focusing lens	2	\$2.00
Protoboard	TX + RX boards	2	\$3.00

Build Steps

Transmitter: ESP32 GPIO drives 2N7000 MOSFET gate through 100Ω resistor. MOSFET drain to laser diode cathode through current-limiting resistor (set for rated current). Laser anode to VCC. Test: toggle GPIO at 1kHz — verify laser blinks with oscilloscope and photodiode. Then implement Manchester encoding: each '1' bit = high-then-low transition, each '0' = low-then-high. Add AES-128 encryption using ESP32's hardware AES engine.

Receiver optics: Mount SFH213 photodiode behind a 25mm focusing lens. Add an 800-820nm optical bandpass filter to reject ambient light. The lens focuses the incoming beam onto the small active area of the diode, increasing the collected photon count. Shield the assembly in a tube painted matte black inside.

Receiver electronics: TIA with 100kΩ feedback (lower than laser mic because we need bandwidth for 1Mbps). Follow with a limiting amplifier or comparator (LM393) that converts the analog signal to clean digital pulses. Feed to ESP32 GPIO input configured for interrupt-on-edge.

Clock recovery + decoding: ESP32 measures the time between each edge transition. Manchester encoding guarantees a transition every bit period. A software DPLL adjusts the sampling point to the center of each bit. Decode Manchester back to binary. Strip framing bytes. Decrypt AES-128. Output plaintext to serial or OLED.

Range test: Start at 10m indoors. Transmit a known message, verify error-free reception. Increase to 50m, 100m, 200m. At each distance, measure BER (bit error rate) by transmitting 10,000 known bits and counting errors. Document BER vs distance. In clear air, you should achieve $<10^{-6}$ BER at 200m with proper alignment.

Interview Questions:

- Why is OOK modulation simpler than PSK for optical links?
- How does Manchester encoding enable clock recovery at the receiver?
- What limits the maximum data rate of this system?
- Why can't this link be intercepted from the side?
- How does atmospheric turbulence affect the link and how would you mitigate it?
- Compare this to RF communication — advantages and disadvantages of each.

PROJECT 3

Laser Rangefinder

Time-of-flight distance measurement with 5ns pulse and 100ps timing resolution

Cost: ~\$25

Difficulty: Advanced

Build Time: 1-2 weekends

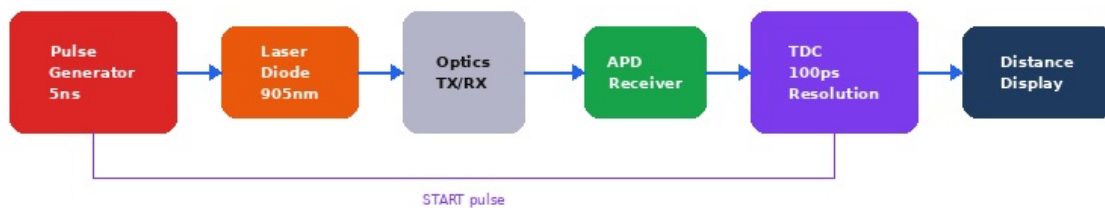
Why This Matters

Every military targeting system, every autonomous vehicle LIDAR, every surveying instrument uses laser time-of-flight ranging. The principle is simple: fire a short laser pulse, start a timer, stop it when the reflection arrives, calculate distance from the round-trip time. The implementation is one of the hardest analog design challenges in EE: generating nanosecond pulses, building a time-to-digital converter with picosecond resolution, and designing a receiver sensitive enough to detect photons reflected from hundreds of meters away.

Why this is the ultimate portfolio project: A working laser rangefinder demonstrates high-speed analog design, precision timing circuits, optical engineering, and signal processing — all in one device. It's the project that makes interviewers stop asking standard questions and start having a technical conversation.

System Architecture

Laser Rangefinder — Time-of-Flight System



5ns pulse → target → reflected photons → APD → TDC measures round-trip time
100ps resolution = 1.5cm distance accuracy | Range: 50-200m | Speed of light timing

How It Works

- **Pulse generator (5ns):** A MOSFET avalanche pulser circuit. A high-voltage supply (100-200V) charges a small capacitor through a resistor. When the MOSFET is triggered, the cap discharges through the laser diode in nanoseconds. The avalanche breakdown of the MOSFET provides the fast switching edge. Pulse width is determined by the capacitance and parasitic inductance — keep leads short.
- **Laser diode (905nm):** A pulsed laser diode rated for high peak power (10-75W) but low average power (duty cycle <0.1%). 905nm is standard for rangefinders because it's eye-safe at the power levels used and atmospheric absorption is low. A collimating lens focuses the beam to <1 mrad divergence.
- **Receiver (APD):** An avalanche photodiode with a narrowband 905nm filter and focusing optics. The APD provides internal gain (x50-200) at a reverse bias of 100-200V. A transimpedance amplifier converts the photocurrent pulse to a voltage pulse. A comparator with adjustable threshold generates the STOP signal for the TDC.
- **Time-to-Digital Converter (TDC):** The heart of the system. The laser trigger pulse is the START signal. The APD comparator output is the STOP signal. The TDC measures the time between them. A TDC7200 IC provides 55ps resolution. Alternatively, build a Vernier delay line TDC from discrete logic — two ring oscillators at slightly different frequencies that achieve interpolated timing down to ~100ps.
- **Distance calculation:** Distance = (time × c) / 2, where c = 3×10⁸ m/s. At 100ps resolution: distance resolution = (100ps × 3×10⁸) / 2 = 1.5 cm. Average multiple measurements to reduce noise and improve resolution further.

Bill of Materials

Component	Part / Spec	Qty	~Cost
905nm pulsed laser diode	SPL PL90_3 or similar	1	\$5.00
APD module	C30902E or AD500-9	1	\$8.00
TDC7200	Time-to-digital converter IC	1	\$4.00
MOSFET	IRF510 (avalanche pulser)	1	\$1.00
HV boost converter	For APD bias + pulser	1	\$3.00
Collimating optics	TX + RX lens assembly	2	\$3.00
ESP32 / STM32	TDC readout + display	1	\$3.00
905nm bandpass filter	Optical, 10nm FWHM	1	\$2.00
Protoboard	With ground plane	1	\$2.00

Build Steps

Avalanche pulser: Charge a 100pF capacitor to 200V through a 10MΩ resistor (slow charge, ~1ms). IRF510 MOSFET gate driven by a fast logic signal (3.3V from MCU through gate driver). When the MOSFET switches, the cap dumps into the laser diode through minimal inductance (short, wide traces). The pulse width is determined by the LC ringing — aim for 5-10ns. Measure with a fast oscilloscope (>500MHz bandwidth).

Laser optics: Mount the pulsed laser diode in a collimating lens tube. Adjust focus for minimum beam divergence at 50m (use a wall as target). The RX optics: a 25mm lens focuses reflected photons onto the APD active area. Mount TX and RX optics side by side with ~3cm separation (bistatic configuration) or coaxially with a beam splitter (monostatic).

Receiver chain: APD biased at its rated voltage (check datasheet — typically 100-200V from a boost converter). TIA with 10kΩ feedback (need bandwidth for 5ns pulse). Follow with a high-speed comparator (LT1016 or ADCMP600) with adjustable threshold. The comparator output is the STOP signal — a clean digital edge triggered by the received photon pulse.

TDC measurement: Connect the laser trigger to TDC7200 START input. Connect the comparator output to STOP input. The TDC7200 communicates via SPI — read the time-interval register. Convert to distance: $d = (\text{time_ns} \times 0.15)$ meters. Average 10 measurements for improved precision. Display on OLED.

Calibration and test: Start with a known distance (measure with tape). Compare rangefinder reading to actual distance. Adjust the fixed time offset (cable delays, processing delays). Then test at multiple distances: 5m, 10m, 20m, 50m, 100m. Plot measured vs actual distance. The slope should be 1.000 ± 0.001 . Document the maximum range (where SNR drops below threshold) and the precision (standard deviation of repeated measurements at fixed distance).

Interview Questions:

- Explain the avalanche pulser circuit — how do you generate a 5ns pulse?
- What determines the minimum and maximum measurable distance?
- How does the TDC achieve picosecond resolution from a MHz clock?
- What limits the range of your system — is it peak power, receiver sensitivity, or optics?
- How would you implement multi-target detection (seeing through glass, for example)?
- What eye safety considerations apply to this design?

PROJECT 4

Encrypted Burst Transmitter

50ms frequency-hopping burst — too fast to DF, too short to jam, encrypted if intercepted

Cost: ~\$10

Difficulty: Intermediate

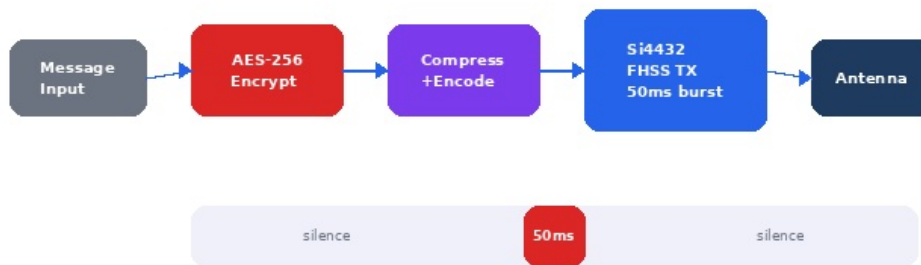
Build Time: 1-2 weekends

Why This Matters

In electronic warfare, the three threats to radio communication are: interception (listening), direction finding (locating the transmitter), and jamming (disrupting the signal). A burst transmitter defeats all three. The message is compressed and encrypted, then transmitted in a single 50ms burst on a pseudo-random frequency. 50ms is too short for most direction-finding systems to get a bearing. Frequency hopping means a jammer doesn't know which frequency to block. AES encryption means even if the burst is recorded, it can't be read. This is the fundamental operating principle of military SINCGARS and HAVE QUICK radios.

System Architecture

Encrypted Burst Transmitter — Transmission Timeline



Entire encrypted message compressed into 50ms burst on random frequency
Too fast to DF | Too short to jam | Encrypted if intercepted | FHSS across band

How It Works

- **Message input:** User types a text message on the ESP32 serial terminal (up to 256 characters). The MCU compresses the text using run-length encoding (saves ~30% on typical messages) and encrypts with AES-256 using a pre-shared key.
- **FHSS sequence generation:** A pseudo-random number generator (seeded with a shared secret) generates a sequence of frequencies across the ISM band (902-928 MHz for US, or 433 MHz band). The TX and RX use the same seed, so they hop to the same frequencies at the same time.
- **Si4432 RF module:** A frequency-agile transceiver that can retune in $<100\mu\text{s}$. The MCU sends the encrypted, compressed packet in a single 50ms burst. During the burst, the frequency hops every 5ms across 10 channels. Total airtime per channel: 5ms — below the threshold of most spectrum analyzers and DF systems.
- **Receiver:** An identical Si4432 + ESP32 module, time-synchronized with the transmitter (either GPS-disciplined clocks or a pre-agreed hop schedule). The receiver hops through the same frequency sequence, collecting 5ms fragments and reassembling the full packet. Decrypt and display.

Bill of Materials

Component	Part / Spec	Qty	~Cost
Si4432 module	433/915MHz FHSS-capable	2	\$4.00
ESP32	TX + RX controllers	2	\$6.00
Quarter-wave antenna	Wire, 82mm (915MHz)	2	\$0.20
OLED display	128x64 I2C	2	\$5.00
LiPo battery	3.7V 500mAh	2	\$4.00
Push button	TX trigger	1	\$0.10

Build Steps

Basic RF link: Wire two Si4432 modules to two ESP32s via SPI. Verify basic communication: send a test message from TX, receive on RX. Measure range with simple continuous transmission — should be 500m+ with wire antenna line-of-sight.

Compression + encryption: Implement RLE compression on the TX ESP32. Then implement AES-256 encryption using the ESP32's hardware AES engine. Pre-share a 256-bit key (flash it to both devices during programming). Verify: transmit an encrypted message, decrypt on the receiver, compare to original.

Frequency hopping: Implement the PRNG-based hop sequence. Both TX and RX use the same seed, so they generate the same frequency list. Program the Si4432 to retune every 5ms across 10 frequencies within the ISM band. The Si4432's frequency resolution is 156Hz, so you can hop precisely. Test: use an RTL-SDR as a spectrum analyzer to observe the signal jumping across frequencies.

Burst assembly: Instead of continuous transmission, the MCU buffers the entire encrypted, compressed message. On button press: the MCU transmits the entire packet in one 50ms burst, hopping 10 times during the burst. Total airtime: 50ms. Then radio silence. The receiver collects all 10 fragments and reassembles.

Time synchronization: For the receiver to hop in sync with the transmitter, both need a shared time reference. Simplest method: both devices start their hop sequence from a pre-agreed time offset after power-on (e.g., aligned to the nearest second using an RTC module). More advanced: GPS-disciplined oscillator for sub-microsecond sync.

Interview Questions:

- How does frequency hopping prevent jamming?
- Why is 50ms too short for direction finding — what DF methods exist and what are their time requirements?
- How do you synchronize the transmitter and receiver hop sequences?
- What's the tradeoff between burst length and message capacity?
- How does your PRNG sequence ensure spectral spreading?
- Compare FHSS to DSSS (direct-sequence spread spectrum) — advantages of each.

PROJECT 5

Passive Night Vision Scope

Image amplification from ambient starlight — zero emission, fully covert

Cost: ~\$30

Difficulty: Advanced

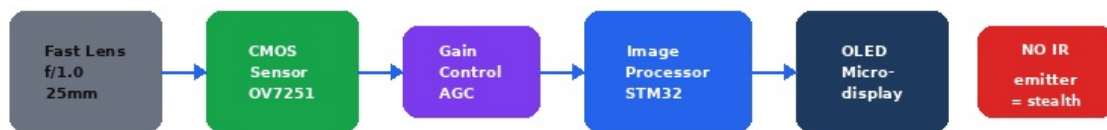
Build Time: 1-2 weekends

Why This Matters

Most commercial 'night vision' devices cheat — they blast IR LEDs and use a camera. This is visible to anyone else with night vision and broadcasts your position. Real military night vision (Gen III image intensifiers) amplifies ambient photons — starlight, moonlight, airglow — without emitting anything. This project builds a digital equivalent: a high-sensitivity CMOS sensor with a fast lens, gain-adjustable analog front end, and OLED microdisplay eyepiece. No IR illuminator. No emission. Fully passive.

System Architecture

Passive Night Vision — Photon Amplification



Fast lens gathers maximum photons → high-sensitivity CMOS → gain-adjusted display
Fully passive: no IR illuminator, no emission, no detectable signature

How It Works

- **Fast lens (f/1.0):** The lens determines how many photons reach the sensor. An f/1.0 lens collects 4x more light than an f/2.0 lens. CCTV lenses with 25mm focal length and f/1.0-f/1.2 aperture are available for \$5-10. This is the single most important component — without a fast lens, no amount of electronic gain will produce a usable image.
- **CMOS sensor (OV7251):** A global-shutter monochrome sensor with high NIR sensitivity. Monochrome sensors are 3-4x more sensitive than color sensors (no Bayer filter absorbing 2/3 of the photons). The OV7251 has 3µm pixels with good quantum efficiency at 850-950nm — where

night sky airglow provides ambient illumination.

- **Gain control (AGC):** The sensor's analog gain is adjustable from 1x to 64x. An AGC loop adjusts the gain based on average image brightness — in darker conditions, gain increases automatically. Additional digital gain (2-8x) on the processor trades noise for visibility in extreme darkness.
- **Image processor (STM32H7 or RPi Zero):** Reads the sensor via MIPI CSI or parallel interface. Applies histogram equalization to maximize contrast. Optionally applies noise reduction (temporal averaging of 2-4 frames). Outputs to OLED microdisplay.
- **OLED microdisplay:** A 0.39" or 0.5" OLED microdisplay viewed through an eyepiece lens. Green phosphor color (matches human scotopic sensitivity) for traditional NV appearance. Refresh rate: 30fps minimum to avoid flicker.

Bill of Materials

Component	Part / Spec	Qty	~Cost
CCTV lens f/1.0	25mm, CS-mount	1	\$6.00
OV7251 module	Monochrome, global shutter	1	\$8.00
RPi Zero 2W	Image processing	1	\$15.00
OLED microdisplay	0.39" SPI, green	1	\$8.00
Eyepiece lens	18mm, from surplus optics	1	\$3.00
LiPo battery	3.7V 2000mAh	1	\$4.00
3D-printed housing	Monocular tube	1	\$2.00

Build Steps

Optical assembly: Mount the f/1.0 lens on the OV7251 camera module using a CS-mount adapter. Verify focus at infinity (point at a distant streetlight at night). The image should be sharp and bright through the viewfinder. If using an RPi camera module instead of bare OV7251, remove the IR-cut filter — this dramatically increases NIR sensitivity.

Sensor configuration: Configure the OV7251 (or RPi camera) for maximum sensitivity: longest exposure time that maintains 30fps (33ms), maximum analog gain (64x), monochrome mode (no demosaicing). Capture test images at night — compare exposure settings systematically to find the optimal gain/exposure tradeoff for your lighting conditions.

Image processing pipeline: On the RPi Zero: capture frames at 30fps. Apply CLAHE (Contrast Limited Adaptive Histogram Equalization) to maximize local contrast. Optionally average 2-4 consecutive frames to reduce temporal noise (at the cost of motion blur). Convert to green-tinted grayscale for traditional NV appearance.

Display output: Route the processed image to the OLED microdisplay via SPI. The display sits at the focal point of the eyepiece lens — the eye sees a virtual image at infinity. Adjust the eyepiece position until the image is sharp and comfortable to view. The total magnification is determined by the ratio of the objective focal length to the eyepiece focal length.

Field test: Test in progressively darker conditions: dusk → street-lit area → moonlit rural area → starlight only. Document the minimum illumination level at which objects are recognizable. Compare to your phone camera at the same location. A well-built passive NV scope should outperform a phone camera by 10-20x in low light due to the fast lens, monochrome sensor, and histogram equalization.

Interview Questions:

- Why does f-number matter more than megapixels for low-light imaging?
- What's the advantage of a monochrome sensor over color for night vision?
- How does CLAHE improve contrast differently from global histogram equalization?
- Why remove the IR-cut filter and what wavelengths does it affect?
- What's the noise floor of your system and how does frame averaging reduce it?
- Compare your digital approach to Gen III analog image intensifiers — advantages and limitations of each.

PROJECT 6

Radar Cross-Section Measurement System

Test stealth shapes and radar-absorbing materials at your desk

Cost: ~\$12

Difficulty: Intermediate

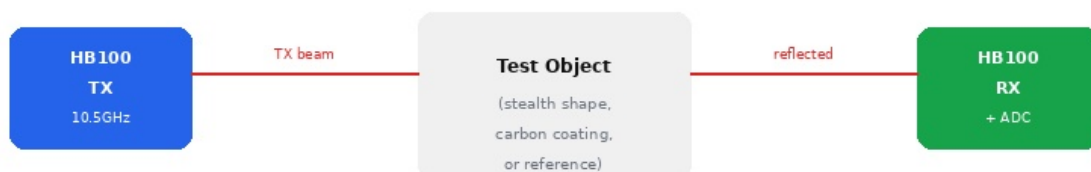
Build Time: 1-2 weekends

Why This Matters

Radar cross-section (RCS) is the measure of how visible an object is to radar. A large RCS means easy to detect; a small RCS means stealthy. Every stealth aircraft, ship, and vehicle is designed to minimize RCS through shape (angling surfaces away from the radar) and materials (absorbing radar energy instead of reflecting it). This project builds a bistatic RCS measurement system using two HB100 modules — one as transmitter, one as receiver — that lets you measure RCS reduction of different shapes and coatings at 10.5 GHz. You're doing what Lockheed does in anechoic chambers, on your desk.

System Architecture

Radar Cross-Section Measurement — Bistatic Setup



TX module illuminates target | RX module measures reflected power
Compare: metal sphere (reference) vs stealth shape vs carbon-coated object
RCS reduction in dB = how invisible the object is to radar

How It Works

- **TX module (HB100):** Transmits continuous wave at 10.525 GHz. Mount facing the target area. The HB100 outputs approximately 15dBm (30mW) — enough for measurements at 1-3m range in a desktop setup. Add a small horn antenna (cut from a tin can) to focus the beam and reduce clutter from reflections off the desk/walls.
- **RX module (second HB100):** A second HB100 used as a receiver. The HB100's mixer output gives a signal proportional to the received 10.5GHz power. Mount at 90° or 180° to the TX (bistatic or forward-scatter configuration). The mixer IF output is amplified and measured.
- **Reference measurement:** First, measure the received power with a known reference target — a metal sphere of known diameter. The RCS of a sphere is $\sigma = \pi \times r^2$ (geometric cross-section). This calibrates the system: you know the relationship between received power and RCS in dBsm.
- **Test targets:** Now replace the reference sphere with your test objects: a flat plate at various angles, a faceted shape (like a stealth aircraft cross-section), a foam shape coated with carbon-loaded paint (RAM — radar absorbing material). Measure received power for each. Calculate RCS relative to the reference. Express the reduction in dB.

Bill of Materials

Component	Part / Spec	Qty	~Cost
HB100 modules	10.525 GHz CW	2	\$8.00
Op-amp	TL072 (IF amplification)	1	\$0.60
ADC module	ADS1115 or Arduino	1	\$3.00
Metal sphere	Steel ball bearing, 25mm	1	\$0.50
Foam shapes	For stealth geometry tests	3	\$1.50
Carbon paint	Graphite/carbon-loaded paint	1	\$3.00
Turntable motor	Slow rotation for angle sweep	1	\$2.00
Protoboard	For op-amp + ADC	1	\$1.50

Build Steps

TX setup: Mount HB100 TX module on a stable platform. Build a simple horn antenna from a tin can (75mm diameter, 100mm length) — solder the HB100's patch antenna centered at the mouth of the can. This provides ~10dB of directional gain, focusing the beam on the target area and reducing wall reflections.

RX setup: Mount the second HB100 at the receive position (1-2m away, same height, facing the target area). Connect the IF output to a TL072 non-inverting amplifier with gain of x100. Follow with a low-pass filter at 100Hz (to average the received power). Feed to ADS1115 ADC read by an Arduino.

Reference calibration: Place a 25mm steel ball bearing at the target position (centered between TX and RX). Record the received power (ADC reading). This is your reference RCS. The theoretical RCS of a 25mm sphere at 10.5GHz: $\sigma = \pi \times (0.0125)^2 = 4.9 \times 10^{-6} \text{ m}^2 = -33 \text{ dBsm}$. All subsequent measurements are relative to this reference.

Test stealth shapes: Cut foam into different shapes: flat plate (high RCS), 45° angled plate (deflects energy away), diamond shape (multiple facets), rounded shape (diffuses energy). Mount each on a slow-rotation turntable (1 RPM). Record received power vs angle. Plot the RCS pattern — you should see dramatic differences between a flat plate (bright return) and an angled shape (greatly reduced return).

Test RAM coatings: Take a flat foam plate (high RCS baseline). Coat one side with carbon-loaded paint (mix graphite powder into acrylic paint). Measure RCS of uncoated side, then coated side. The carbon paint should absorb some radar energy, reducing the RCS by 3-10dB depending on coating thickness and carbon loading. Experiment with different carbon concentrations to find optimal absorption.

Interview Questions:

- What is radar cross-section and what determines it?
- Why use a sphere as a calibration reference?
- How does object shape affect RCS — explain the principle behind stealth geometry?
- What physical mechanism causes carbon-loaded paint to absorb radar energy?
- What's the difference between bistatic and monostatic RCS measurement?
- How would you build a proper anechoic chamber to reduce measurement clutter?

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